Advanced Computer Graphics (ACG)

Theory Session 3 – Ray Casting and Recursive Ray Tracing

2026-2027

Pompeu Fabra University

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Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Class Outline

- Accelerating Ray-Scene Intersections
  - Bounding Volumes
  - Ray-Tracing Acceleration Structures
- Ray Casting – Technical Insights
  - Point Light Sources and Ray Reflection at a Surface
  - Phong Reflection Model
- Recursive Ray Tracing – Technical Insights
  - Specular Reflection and Refraction

Ray Tracing: Performance Challenges (Last Class)

- Basic ray-scene intersection:
  - Test intersection with all the scene objects
- How costly is this?
  - #intersection tests $\geq$ #pixels ($\approx 10^6$) x #triangles (?)
  - How many triangles on a complex scene?



San Miguel's Scene, 10.7M triangles

Jun Yan, Tra

Plant Ecosystem, 20M triangles



Deussen et al; Pharr & Humphreys, PBRT

Ray Tracing: Performance Challenges (Last Class)

- Basic ray-scene intersection:
  - Test intersection with all the scene objects
- How costly is this?
  - #intersection tests $\geq$ #pixels $(\sim 10^6)$ x #triangles $(\sim 10^6)$
- Problem!
  - Extremely computing-intensive approach
  - Needs improvement
  - **Suggestions?**

Bounding Volumes

Bounding Volumes

- Key idea:
  - Define a simple volume containing the object (i.e., the mesh)
  - Example of bounding volumes are spheres or boxes
  - To test intersection with the mesh:
    - First test intersection with the bounding volume
    - If the ray does not intersect the volume, then it does not intersect the mesh

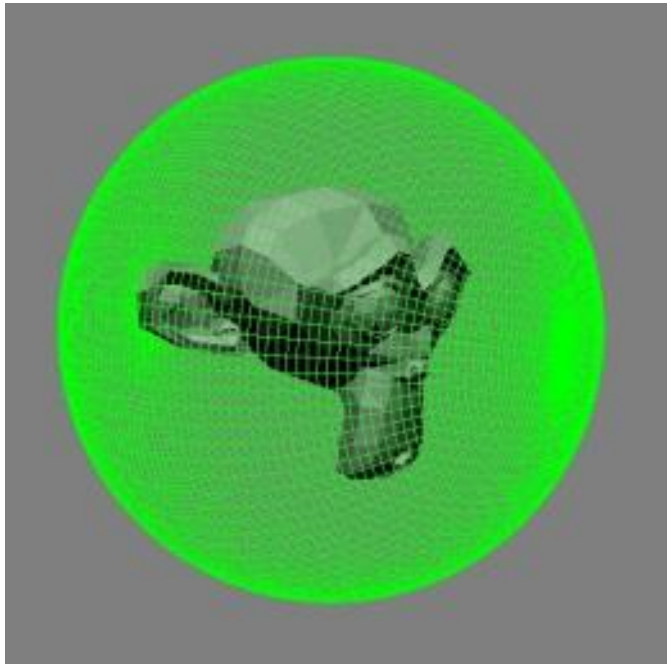
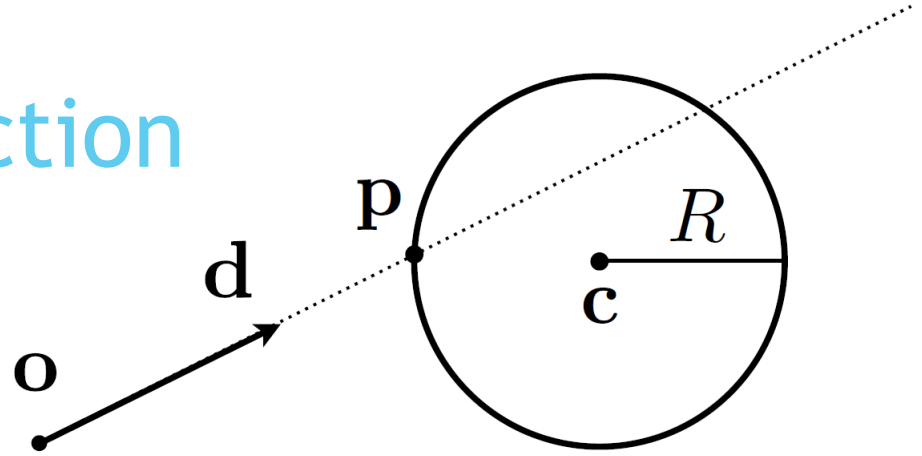


Image courtesy of Sarah Noonan

Ray-Sphere Intersection



Ray: $\boldsymbol{r}(t) = \boldsymbol{o} + t\boldsymbol{d}, \quad 0 \leq t < \infty$

Sphere: $\boldsymbol{p} : (\boldsymbol{p} - \boldsymbol{c})^2 - R^2 = 0$

Solve to **find** $\boldsymbol{t}$:

$(\boldsymbol{o} + t\boldsymbol{d} - \boldsymbol{c})^2 - R^2 = 0$

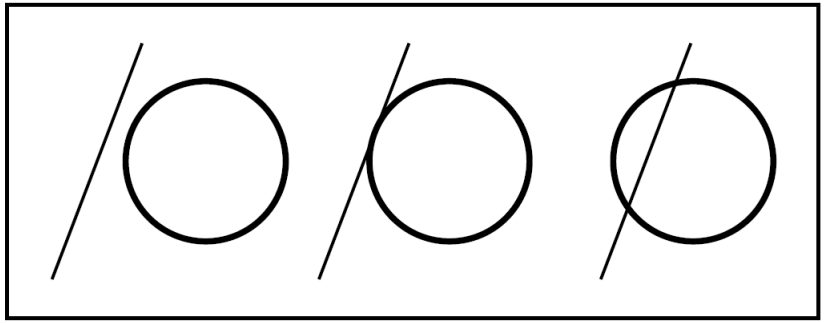
$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$at^2 + bt + c = 0$, where:

$a = \boldsymbol{d} \cdot \boldsymbol{d}$

$b = 2(\boldsymbol{o} - \boldsymbol{c}) \cdot \boldsymbol{d}$

$c = (\boldsymbol{o} - \boldsymbol{c}) \cdot (\boldsymbol{o} - \boldsymbol{c}) - R^2$



Images from Berkeley CS184/284A

Exercise 1: Bounding Sphere

Consider the mesh associated to a solid tetrahedral object whose vertices are at the coordinates $A = (4, 4, 4)$, $B = (7, 4, 4)$, $C = (4, 8, 4)$ and $D = (4, 4, 9)$. A camera is located at the position $E = (2, 2, 2.5)$.

1) Define a bounding sphere for this object.

2) Consider a ray with origin E and direction $d = (0.70711, 0.70711, 0)$; decide about the eventual intersections with the bounding sphere defined.

Bounding Volumes: Limitations

- The efficiency of the bounding volume directly depends on how well it fits to the enclosed geometry
  - The larger the volume, the more likely a given ray intersects the volume ➔ Test intersection with all enclosed triangles
  - Several types of bounding volumes: bounding sphere, bounding box, etc.

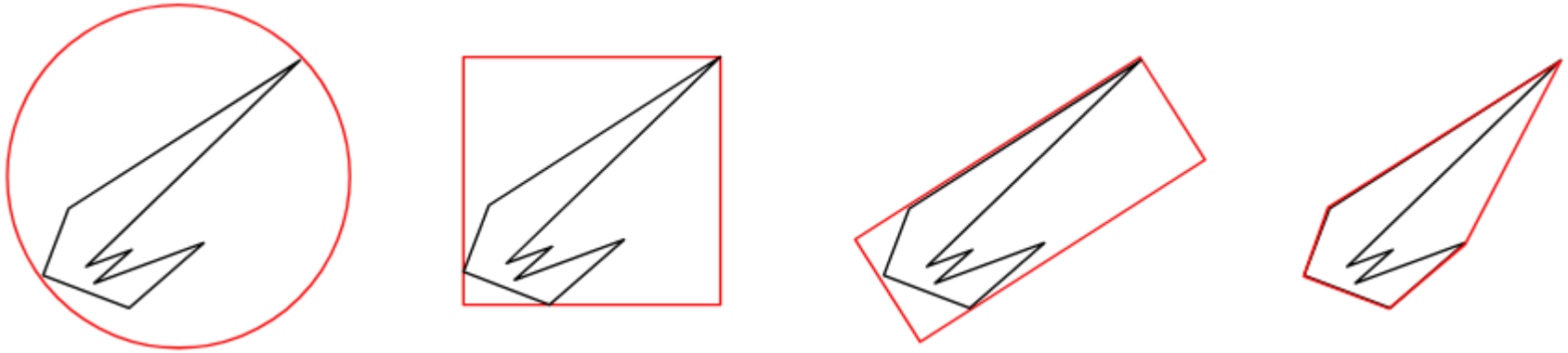
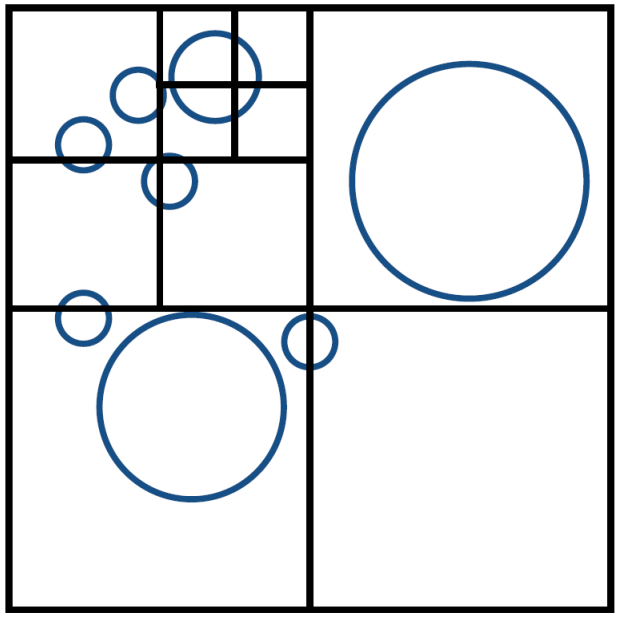


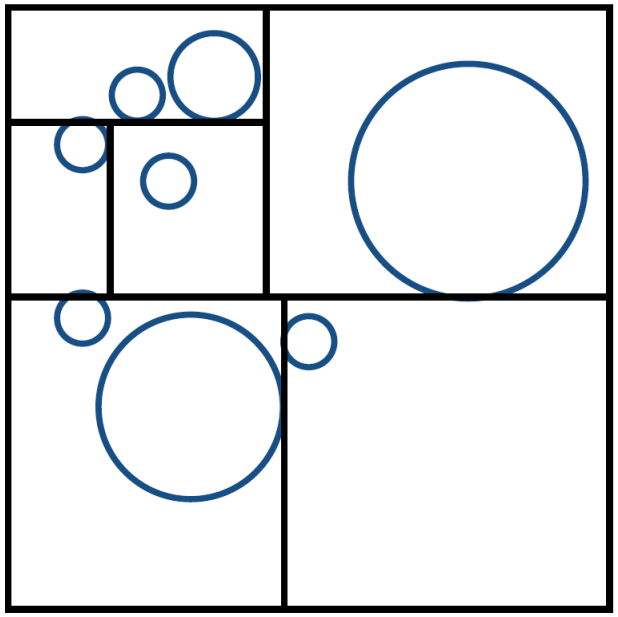
Image from ncollide.org

- How could we do better?

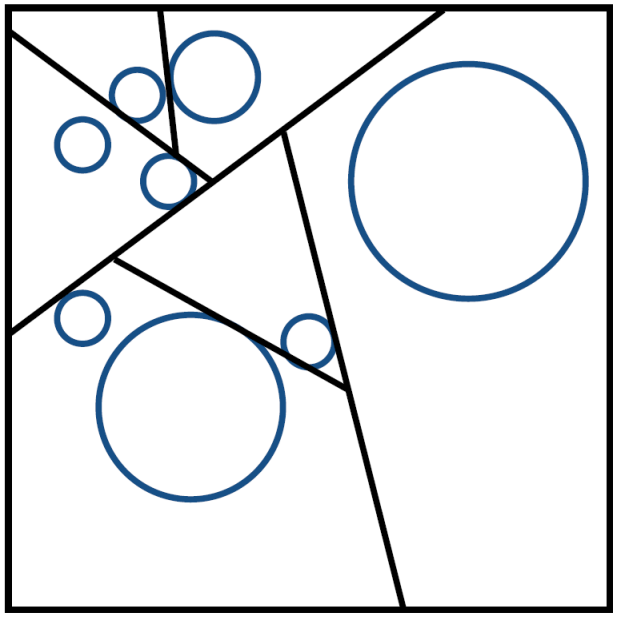
Acceleration Structures for Ray Tracing



Oct-Tree



KD-Tree



BSP-Tree

Interesting video

Images from Berkeley CS184/284A

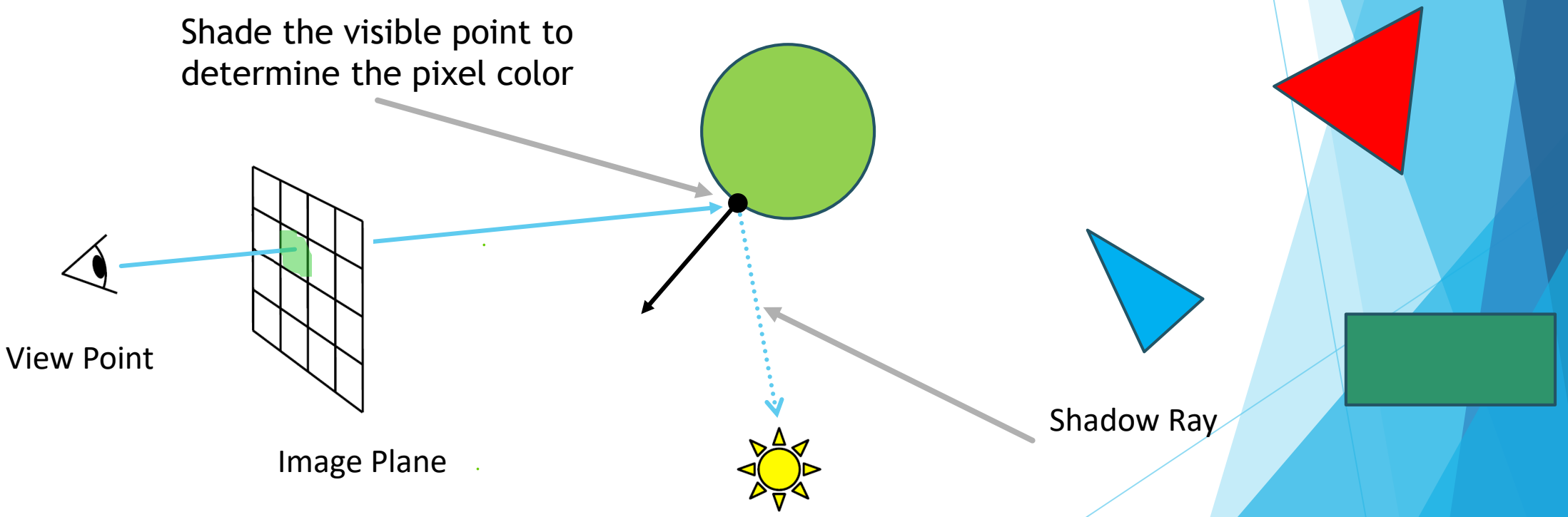
Ray Casting

Point Light Sources

Ray Reflection at a Surface

Phong Reflection Model

Ray Casting – Shading (Local)



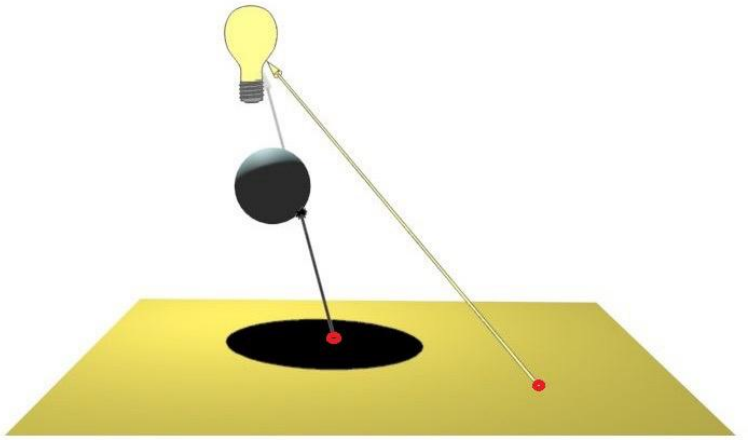
Ray Casting Requirements

- Compute ray-scene intersection (✓)
  - Determine visible points (aka shading points)
  - Determine occlusion of the light sources

- Shade ("paint") the points lying on non-specular surfaces (✖)
  - Compute how much light arrives at the shading point
    - Implies modelling the scene light sources
  - Compute how much of this light is reflected towards the viewer
    - Implies modelling the material associated with that shading point

Video

Point Light Sources



- Simplest type: diffuse point light sources
  - "*Diffuse*" because light is equally emitted in all directions
  - "*Point*" because there are modelled as an infinitely small point (no area)
  - Characterized by a 3D position $\boldsymbol{p}_L$ and an intensity $\boldsymbol{I}_L$
  - Do not exist in the real world (so, not realistic)

- Advantages
  - Simple to model
  - Its visibility can be sampled using a single ray ➔ leads to faster images

- Disadvantage
  - Do not exist in the real world ➔ not realistic (e.g., hard shadows)

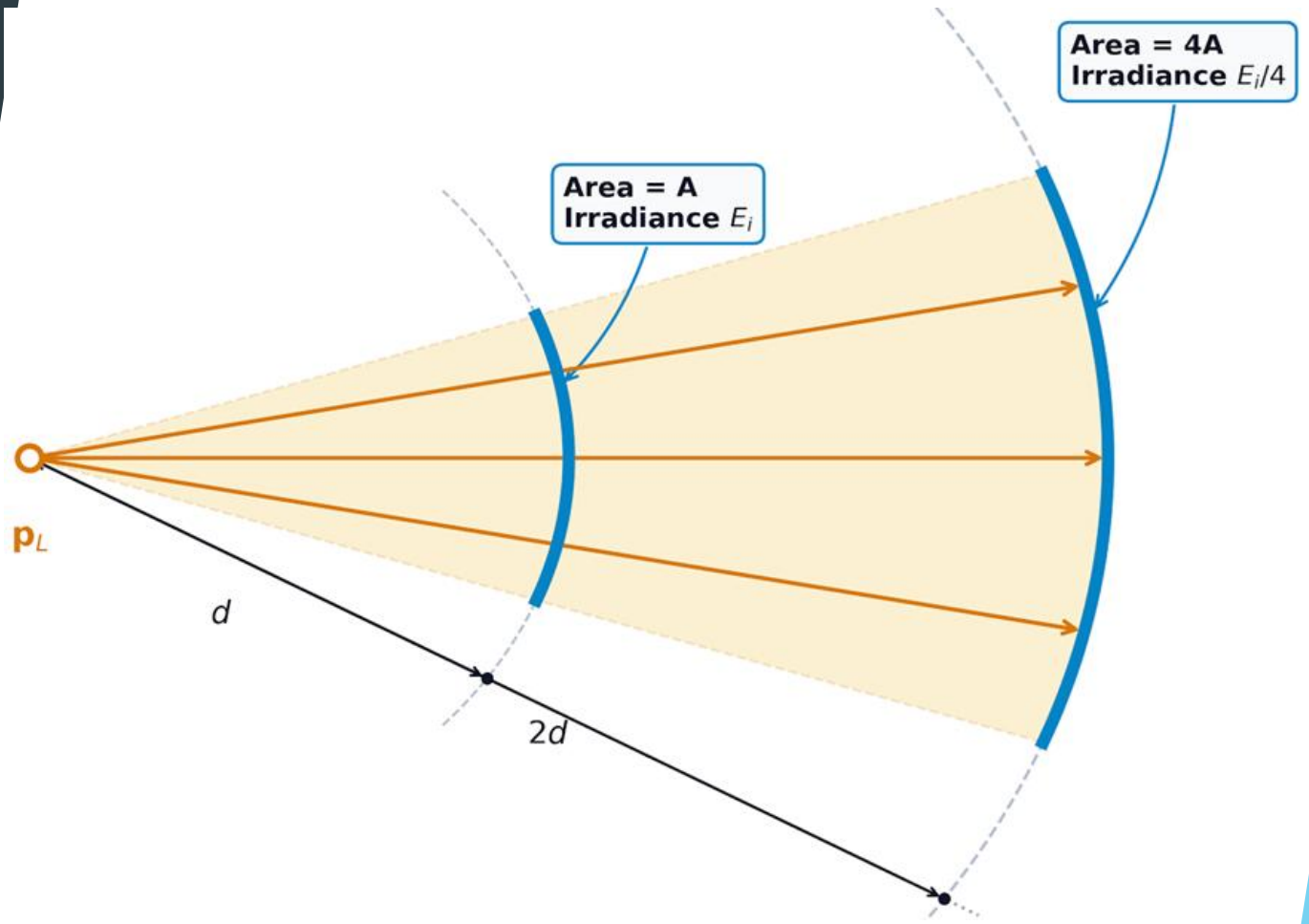
Point Light Sources

- Incoming **direct** irradiance $\mathrm{E}_i(\boldsymbol{p})$ at a shading point $\boldsymbol{p}$ (perpendicular to light direction) due to a light source $\boldsymbol{L}$ with position $\boldsymbol{p}_L$:

$$\mathrm{E}_i(\boldsymbol{p}) = \frac{\boldsymbol{I}_L}{d^2} \text{ , where } d = \|\boldsymbol{p}_L - \boldsymbol{p}\|$$

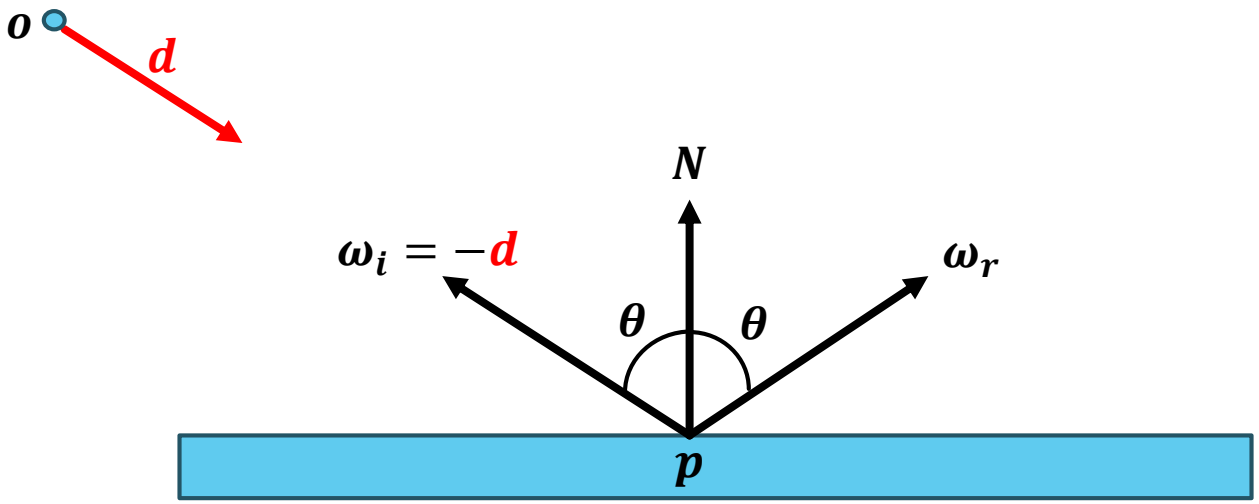
Note on notation:

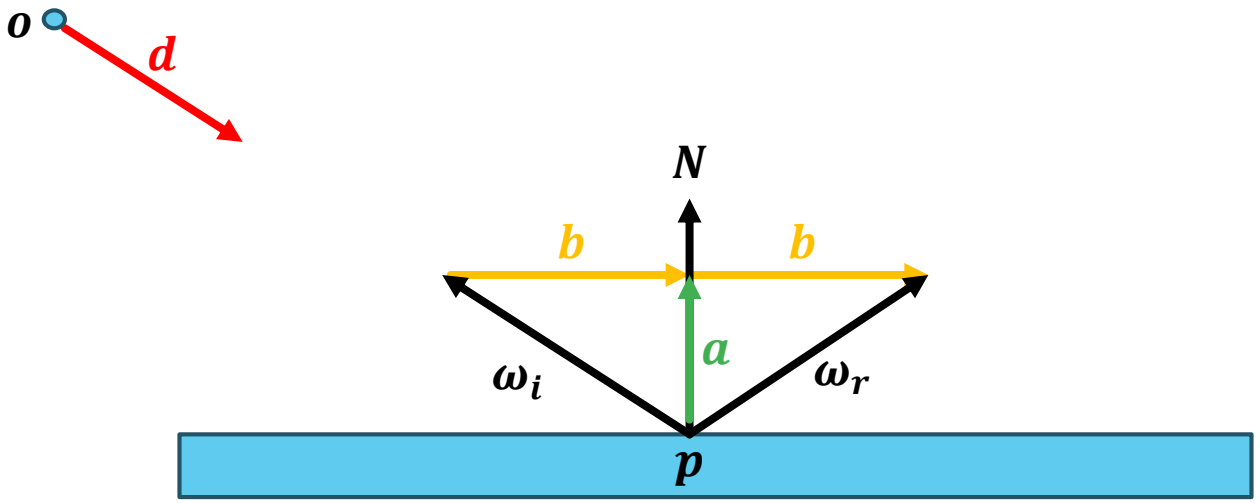
E_i ➔ Incident irradiance



Law of Reflection

- Governs the reflection at perfect specular materials (e.g., mirror)
- The reflection direction depends on the direction of incidence and on the surface orientation
- **Problem set up**: let us consider the ray $\boldsymbol{r}(t) = \boldsymbol{o} + t\boldsymbol{d}$. Let the point p (intersected by the ray) have a normal $\boldsymbol{N}$. Let $\boldsymbol{\omega_i} = -\boldsymbol{d}$ (by convention $\boldsymbol{\omega_i}$ points away from p).
- **Objective**: compute the perfect specular reflection $\boldsymbol{\omega_r}$ such that $\boldsymbol{\omega_i} \cdot \boldsymbol{N} = \boldsymbol{\omega_r} \cdot \boldsymbol{N} = \cos\boldsymbol{\theta}$

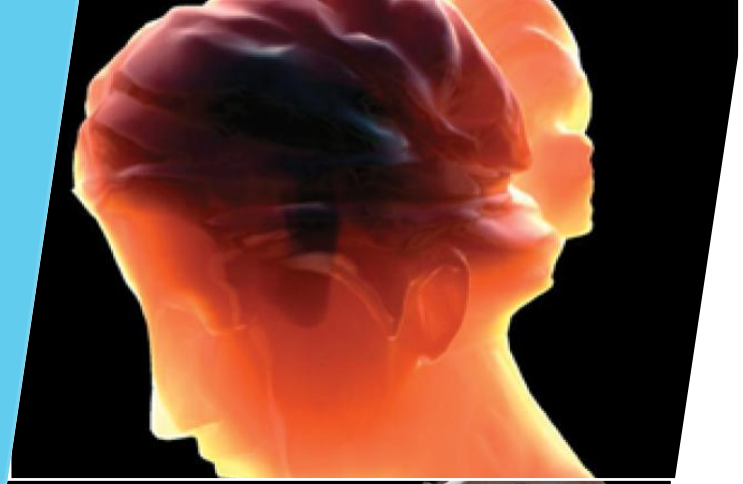


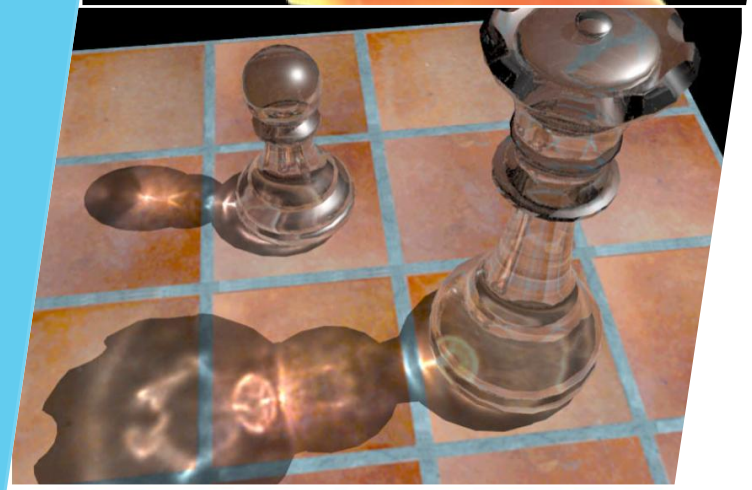


- **Observations:**
  - If we define the vectors $\boldsymbol{a}$ (green) and $\boldsymbol{b}$ (yellow), then $\boldsymbol{\omega_r = a + b}$
  - a is the projection of $\boldsymbol{\omega_i}$ onto $\boldsymbol{N}$ such that $\boldsymbol{a = \mathrm{N} \cos \theta = (\omega_i \cdot N)\, N}$
  - It is easy to see that $\boldsymbol{\omega_i + b = a}$, from where $\boldsymbol{b = a - \omega_i = (\omega_i \cdot N)\, N - \omega_i}$

- We can now replace $\boldsymbol{a = (\omega_i \cdot N)\, N}$ and $\boldsymbol{b = (\omega_i \cdot N)\, N - \omega_i}$ in the equation $\boldsymbol{\omega_r = a + b}$, yielding:

$$\boldsymbol{\omega_r = (\omega_i \cdot N)\, N + (\omega_i \cdot N)\, N - \omega_i = 2\,(\omega_i \cdot N)\, N - \omega_i}$$



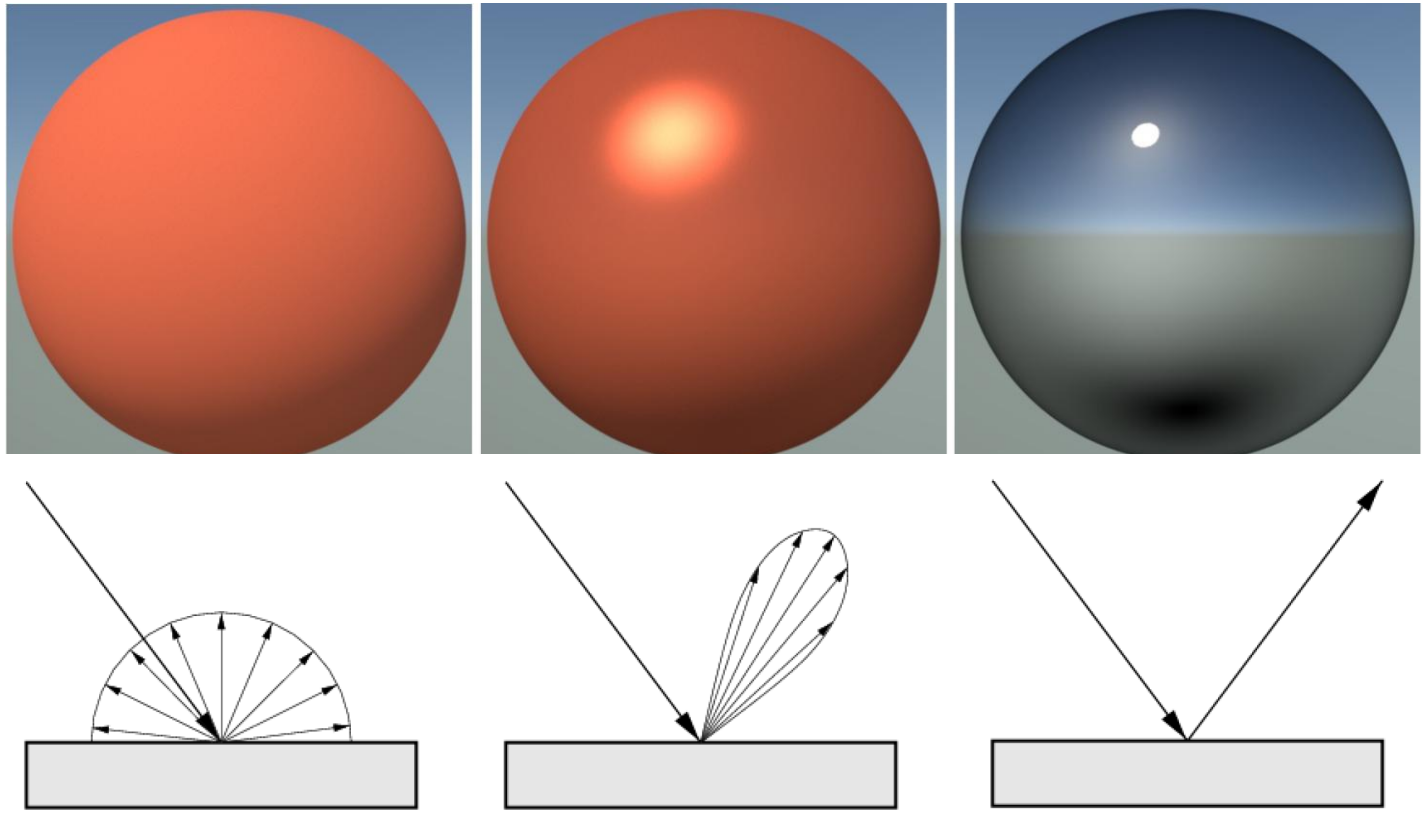




Light Effects

- There are different types of light-matter interaction
  - Reflections at the surface
    - Specular (mirror), diffuse or glossy
  - Refraction
    - When light changes propagation medium
    - Caustics
  - Sub-surface scattering

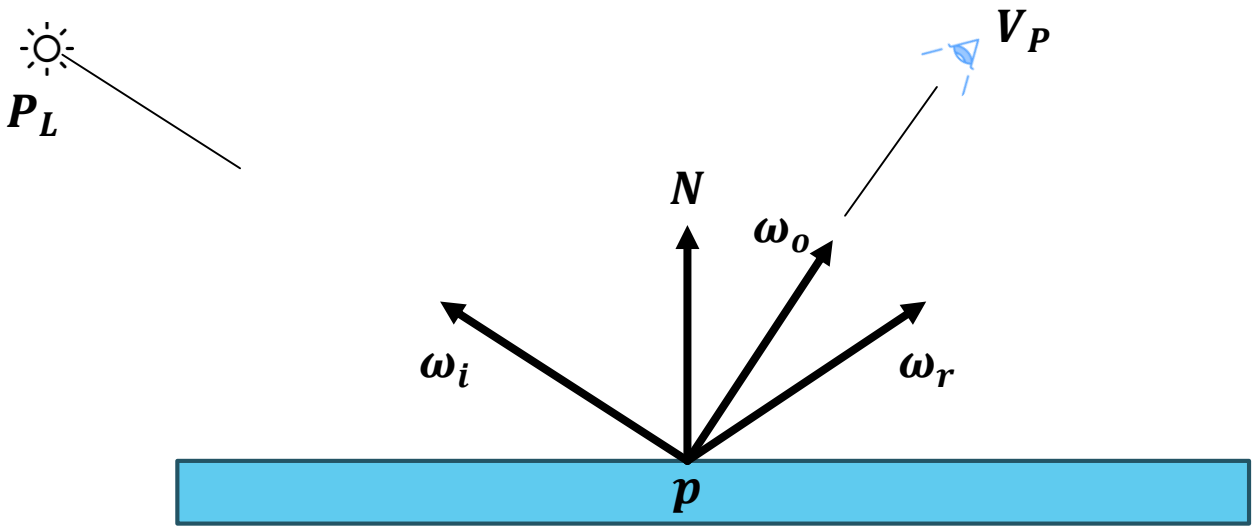
Specular, Glossy and Diffuse Reflections



(a) Perfect diffuse reflection (b) Glossy specular reflection (c) Perfect specular reflection

Phong Reflection Model

- Problem set-up: let us consider the following configuration

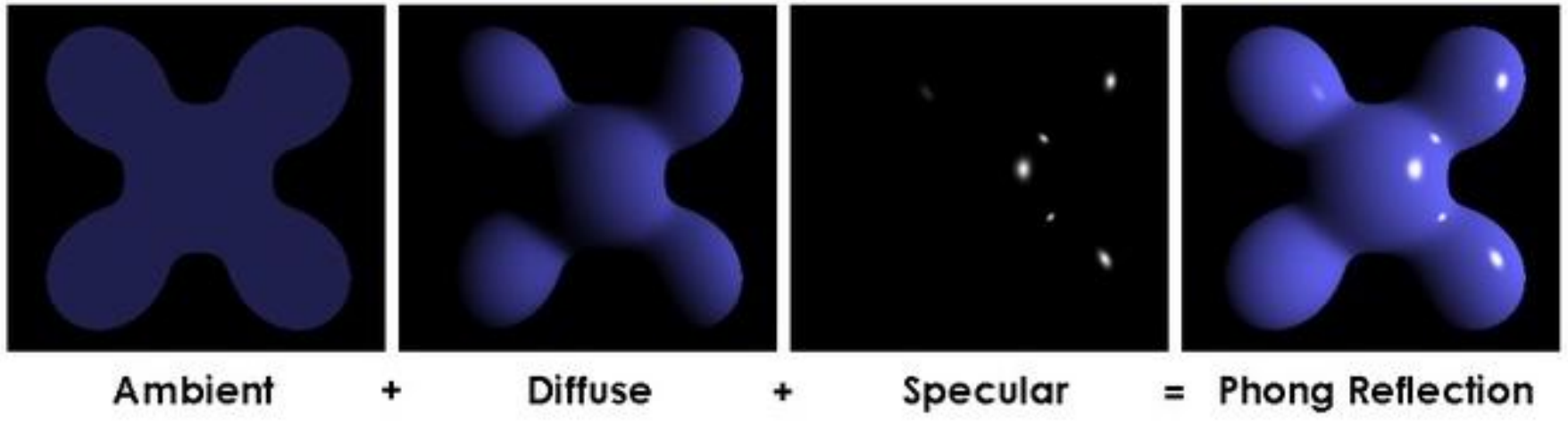


Recall: $\omega_r = 2(\omega_i \cdot N)\, N - \omega_i$

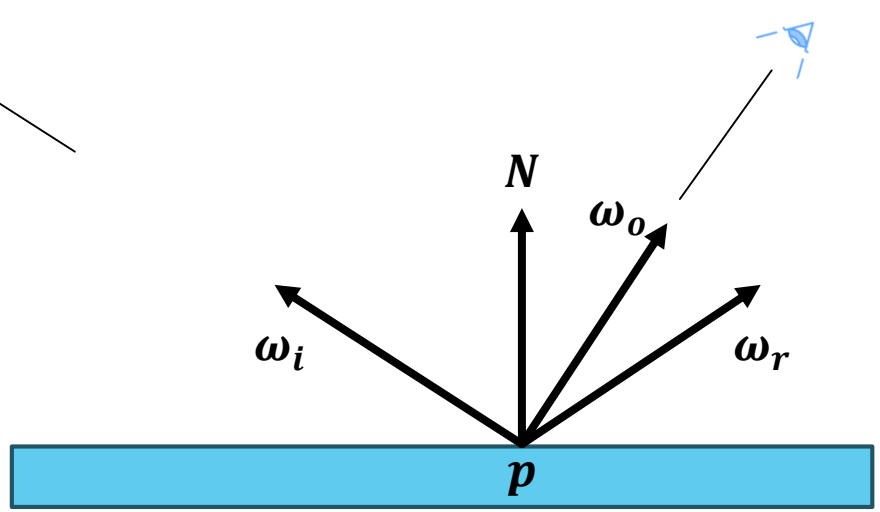
- Goal determine the appearance of point p as seen from the viewpoint V_P

Phong Illumination (Demo)

- Empirical model of the reflection of light at a surface (video)
  - **Direct light** reflection is modeled through a combination of **diffuse** and **specular** reflections
  - **Indirect light** reflection is approximated through an **ambient term** (constant across all the scene)

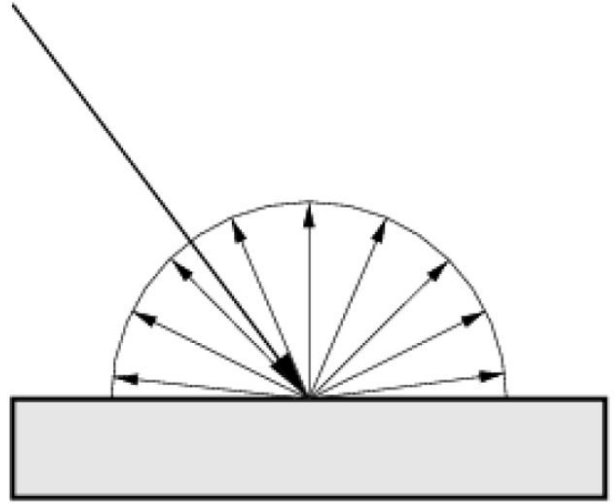


Diffuse Reflection



- Diffuse reflection
  - Incident light is equally scattered in all possible directions

- The amount of (diffusely) reflected light depends on:
  - The intensity of the light source (I_L) and its distance (d) to the shading point
  - The angle of incidence of light ($\boldsymbol{\omega_i} \cdot \boldsymbol{N}$)
  - The diffuse coefficient (k_d). It captures the fraction of incoming light that is reflected diffusely, and determines the diffuse color of the material
  - Does not depend in the viewing direction $\boldsymbol{V}$

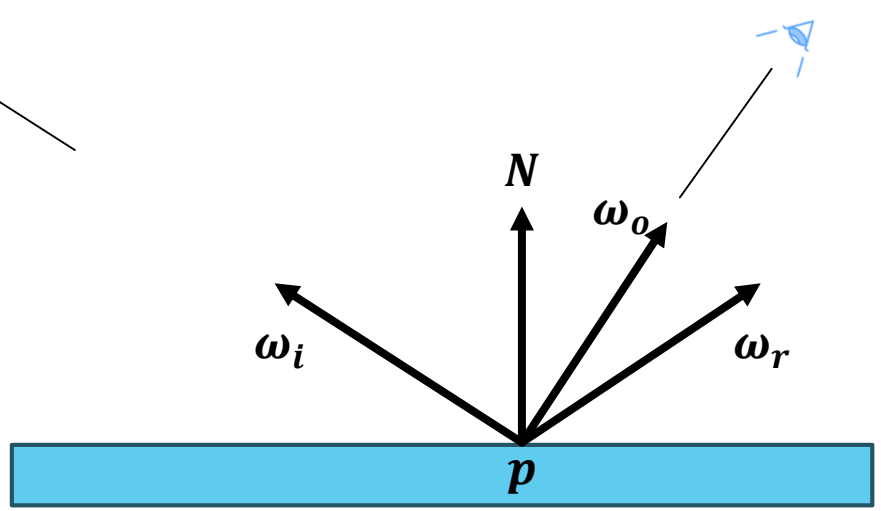


Recall: $\mathrm{E}_i(\boldsymbol{p}) = \frac{\boldsymbol{I_L}}{d^2}$

$$\mathrm{L}_o^{(d)} = \mathrm{E}_i(\boldsymbol{p})(\boldsymbol{\omega_i} \cdot \boldsymbol{N})\, k_d$$

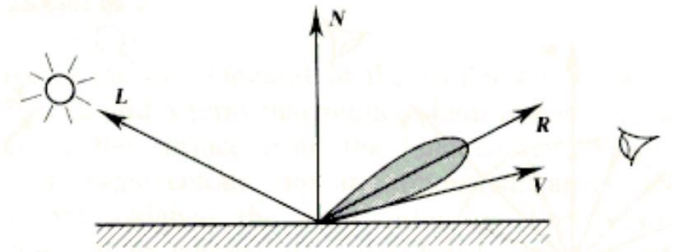
Incident light weighted by the angle of incidence

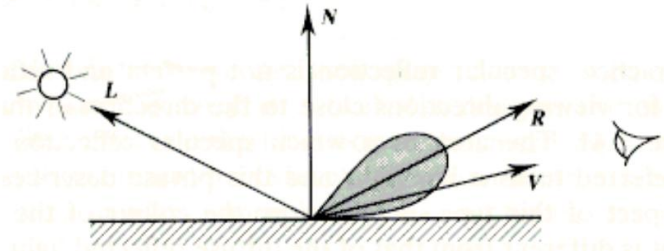
Specular (aka Glossy) Reflection



- Specular reflection
  - Incident light is mainly scattered around the perfect reflection direction $\boldsymbol{\omega_r}$

- The amount of specularly reflected light depends on:
  - The amount of light arriving at $\boldsymbol{p}$ from the light source
  - The angle of incidence of light $(\boldsymbol{\omega_i} \cdot \boldsymbol{N})$
  - Then angle between the viewing direction $(\boldsymbol{\omega_o})$ and the perfect reflection direction $(\boldsymbol{\omega_r})$
  - The specular coefficient (k_s) which models the fraction of incident light which is reflected specularly
  - The shininess coefficient (s, aka specular exponent or shininess exponent)





$$\mathrm{L}_o^{(s)} = \underbrace{\mathrm{E}_i(\boldsymbol{p})(\boldsymbol{\omega_i} \cdot \boldsymbol{N})}\, k_s(\boldsymbol{\omega_r} \cdot \boldsymbol{\omega_o})^s$$

Incident light weighted by the angle of incidence

Recall: $\mathrm{E}_i(\boldsymbol{p}) = \frac{\boldsymbol{I_L}}{d^2}$

Ambient Term

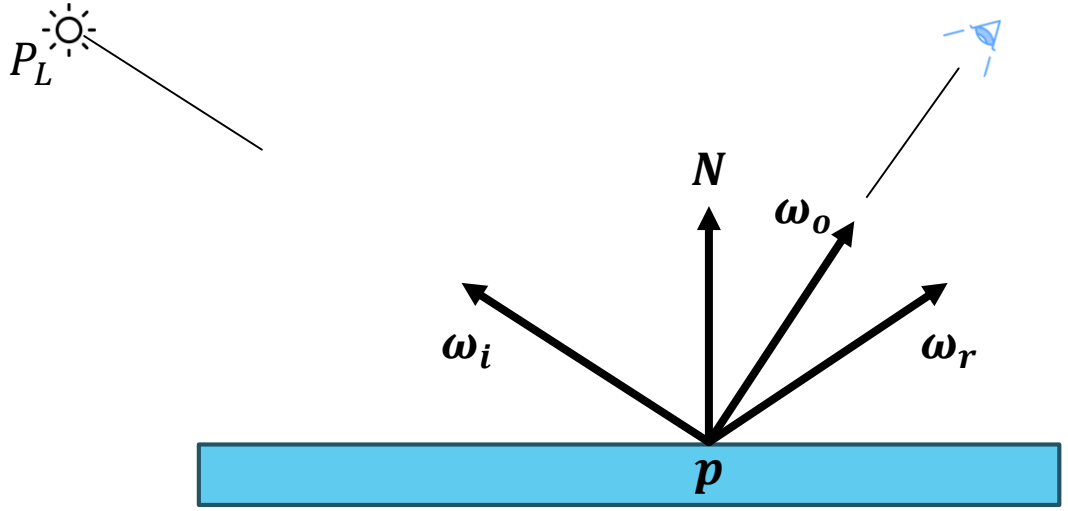


From left to right: direct (local), indirect (global) and full illumination

- Ambient component
  - Represents the light which arrives at the shading point after one or more reflections on the scene objects
  - Usually a user-provided per-scene parameter (I_a)
  - Strong simplification:
    - We assume that the same amount of indirect light arrives at all scene points
    - Leads to lack of realism… but it's simple to implement and very fast to compute
    - k_a is the ambient coefficient of the material

$$\mathrm{L}_o^{(a)} = k_a \, I_a$$

Putting it all Together



- The outgoing light (L_o) at point $\boldsymbol{p}$ towards direction $\boldsymbol{\omega_o}$ is given by:

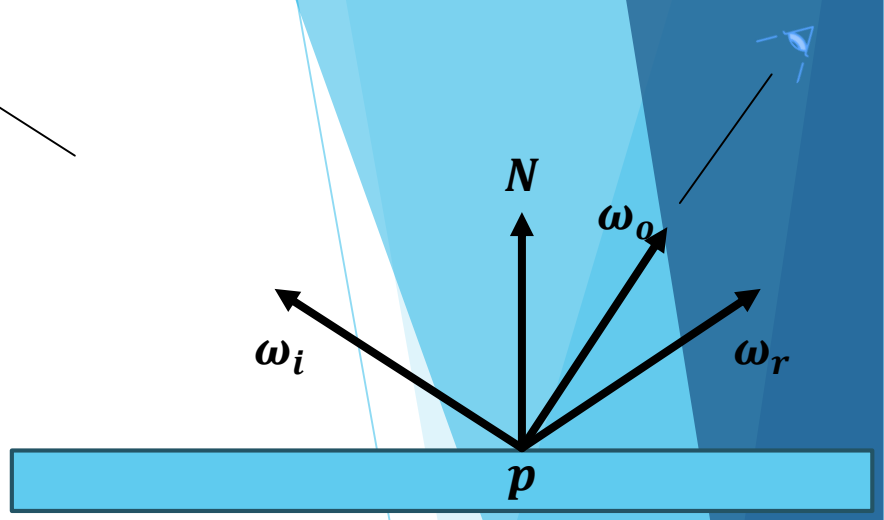
$$L_o(\boldsymbol{\omega_o}) = V(\boldsymbol{p}, P_L)\,(\underbrace{E_i(\boldsymbol{p})(\boldsymbol{\omega_i}\cdot\boldsymbol{N})\,k_d}_{L_o^{(d)}} + \underbrace{E_i(\boldsymbol{p})(\boldsymbol{\omega_i}\cdot\boldsymbol{N})\,k_s\,(\boldsymbol{\omega_r}\cdot\boldsymbol{\omega_o})^s}_{L_o^{(s)}}) + \underbrace{k_a I_a}_{L_o^{(a)}}$$

Where:

- k_d, k_s and k_a are **[R,G,B]-tuples**, and $0 \leq R, G, B \leq 1$
- The result $L_o(\boldsymbol{\omega_o})$ is also a color, and represents the appearance of point $\boldsymbol{p}$ as seen from the viewer's position
- $V(\boldsymbol{p}, P_L)$ is the **visibility function** and yields 1 if P_L is visible from $\boldsymbol{p}$, and 0 otherwise

- **IMPORTANT**: In practice, need to distinguish between colors, 3D vectors, 3D points
  - All of them can be represented by vectors in C++
  - Mathematical operations for each type are different (e.g., color multiplication is performed independently per canal)

Phong Reflection Model



- Separating the terms related to incident light and using:

$$f_r(\boldsymbol{\omega_i}, \boldsymbol{\omega_o}) = k_d + k_s(\boldsymbol{\omega_r} \cdot \boldsymbol{\omega_o})^s$$

we can re-write $\mathrm{L}_o(\boldsymbol{\omega_o})$ as:

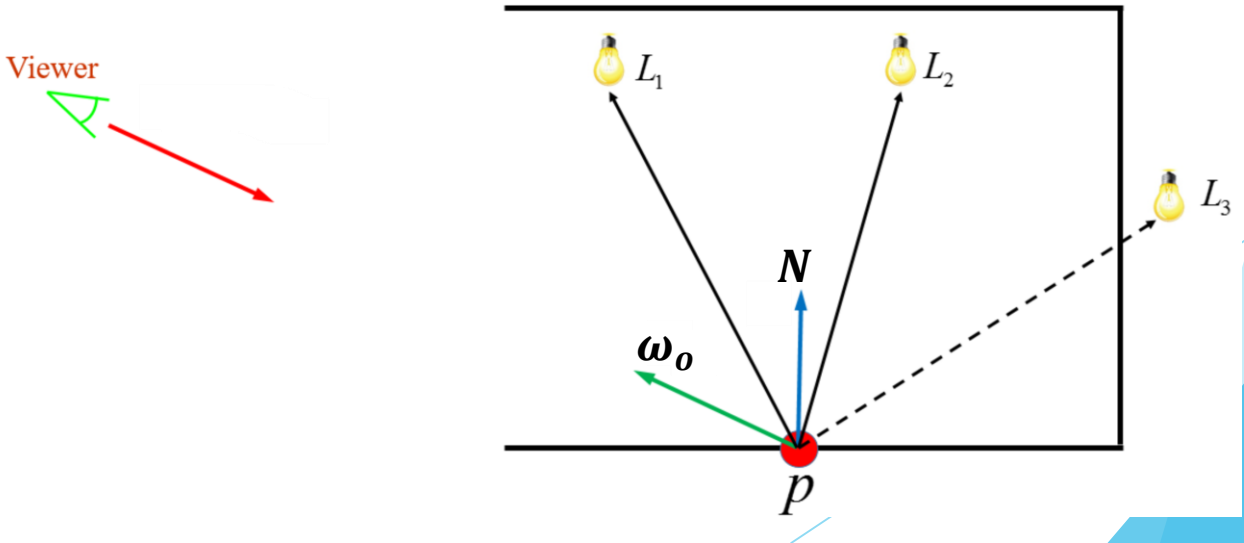
$$\mathrm{L}_o(\boldsymbol{\omega_o}) = \mathrm{V}(\boldsymbol{p}, P_L)\, \mathrm{E}_i(\boldsymbol{p})(\boldsymbol{\omega_i} \cdot \boldsymbol{N})\, f_r(\boldsymbol{\omega_i}, \boldsymbol{\omega_o}) + k_a I_a$$

where $f_r(\boldsymbol{\omega_i}, \boldsymbol{\omega_o})$ is the "*Phong Reflection Model*" and yields the fraction of light arriving from $\boldsymbol{\omega_i}$ which is reflected towards $\boldsymbol{\omega_o}$

Phong Reflection Model with Multiple Light Sources

- If there are J point light sources, then we must account for the contribution of each light source

$$\mathrm{L}_o(\boldsymbol{\omega_o}) = k_a I_a + \sum_{\boldsymbol{j}=1}^{J} \mathrm{V}\left(\boldsymbol{p}, P_{L_{\boldsymbol{j}}}\right) \mathrm{E}_{i_{\boldsymbol{j}}}(\boldsymbol{p})\left(\boldsymbol{\omega_j} \cdot \boldsymbol{N}\right) f_r(\boldsymbol{\omega_j}, \boldsymbol{\omega_o})$$



Exercise 2: Phong Material

Consider a scene containing a point P with coordinates $P = (0, 0, 0)$, and a normal at the surface where P lies such that $N = (0, 0, 1)$. The surface material at P is of type Phong, with a diffuse coefficient $k_d = (1, 0, 0)$, a specular reflection coefficient $k_s = (0, 0.9, 0.9)$, and a shininess coefficient $s = 20$. Assume that the ambient coefficient of the scene is $k_a = (1, 0, 0)$, and that the scene contains a point light source with position $P_L = (4, 0, 3)$ and intensity $I_L = (2500, 2500, 2500)$.

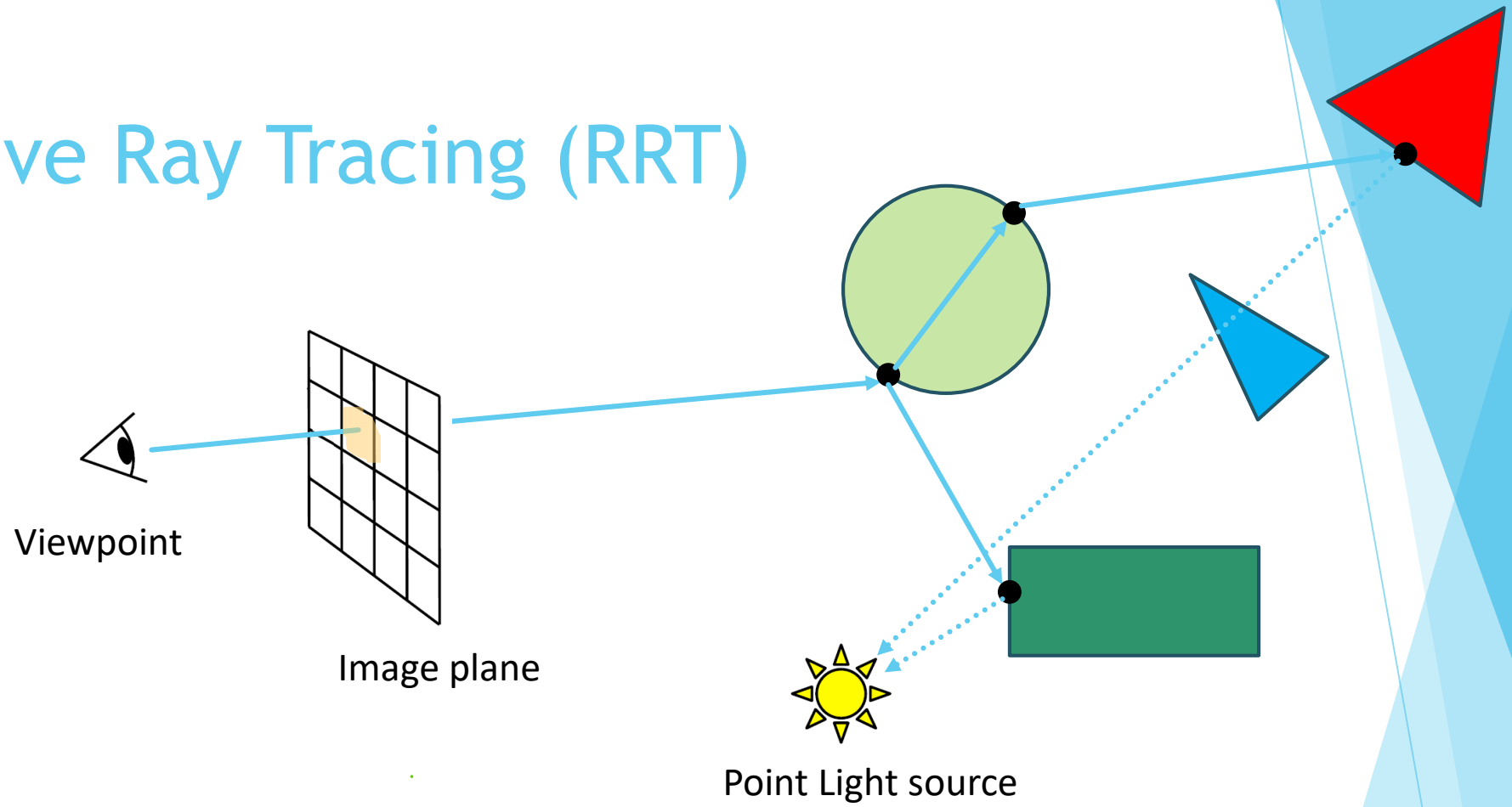
Questions: Knowing that the scene has an ambient light $I_a = (2.5, 2.5, 2.5)$, and considering a viewer located at the position $P_V = \left(-\frac{8}{5}, 0, \frac{6}{5}\right)$, compute:

1) The amount of light that arrives at point P from the light source;
2) The amount of light reflected at point P due to the diffuse reflection.
3) The reflected light due to specular reflection at P towards the viewer
4) The total light reflected at P towards the viewer.

Recursive Ray Tracing

Specular Reflection and Refraction

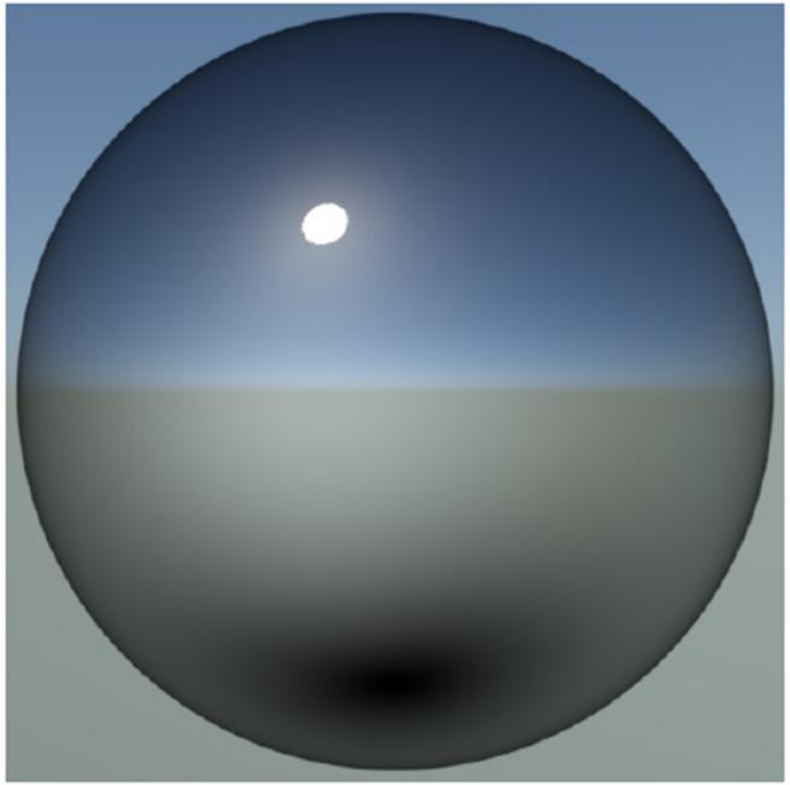
Recursive Ray Tracing (RRT)



- Recursively trace secondary rays until a non-specular surface is hit
- At each intersection point, evaluate light sources visibility (no contribution if occluded)
- Final pixel color results from all light contributions along the paths
- Allows simulating specular reflections and transmissions

RRT Requirements

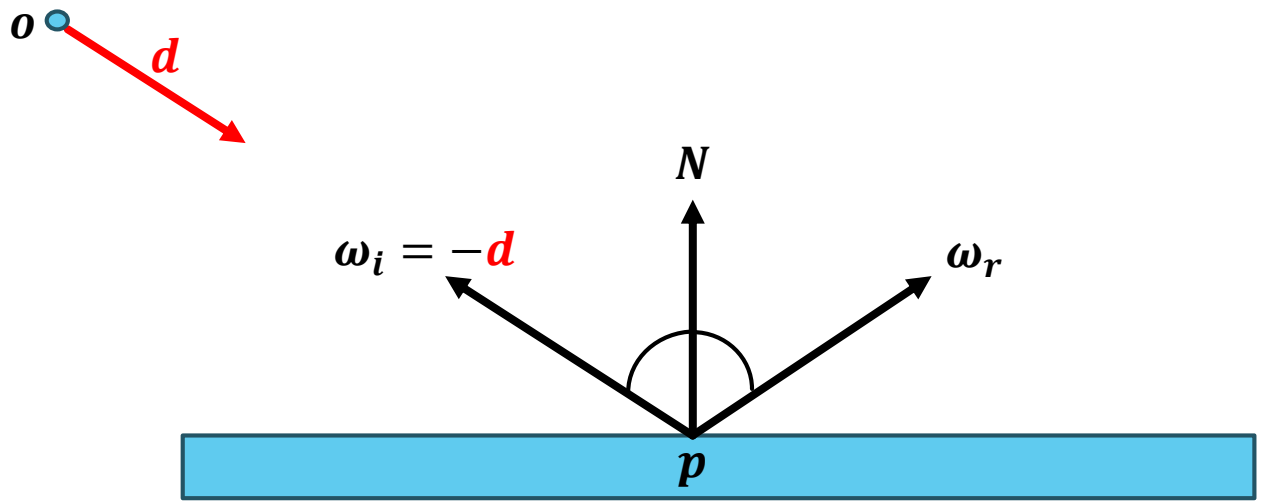
- Compute ray-scene intersection (OK)
  - Determine visible points (aka shading points)
  - Determine occlusion of the light sources
- Shade ("paint") the points lying on non-specular surfaces (OK)
  - Compute how much light arrives at the shading point
  - Compute how much of this light is reflected towards the viewer
- **Reflect** and **refract** incident rays
  - Required for specular materials



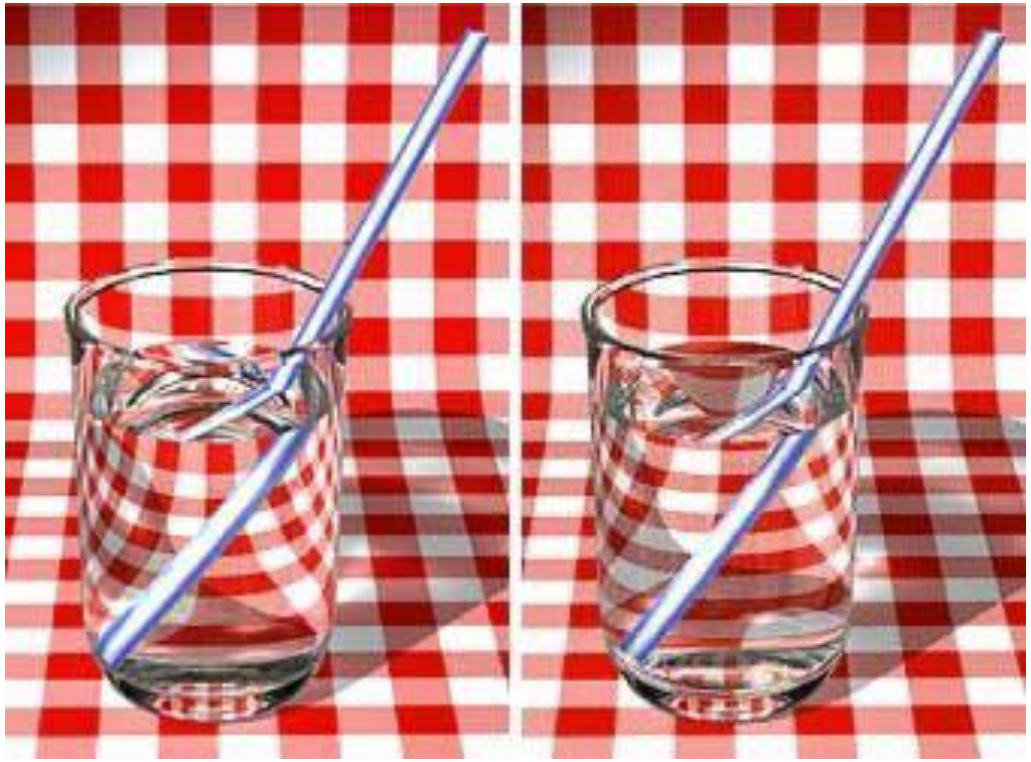
Specular Reflection

Specular Reflection

- Works as a perfect mirror
- Reflect the incident ray at the intersection point using the *Law of Reflection*

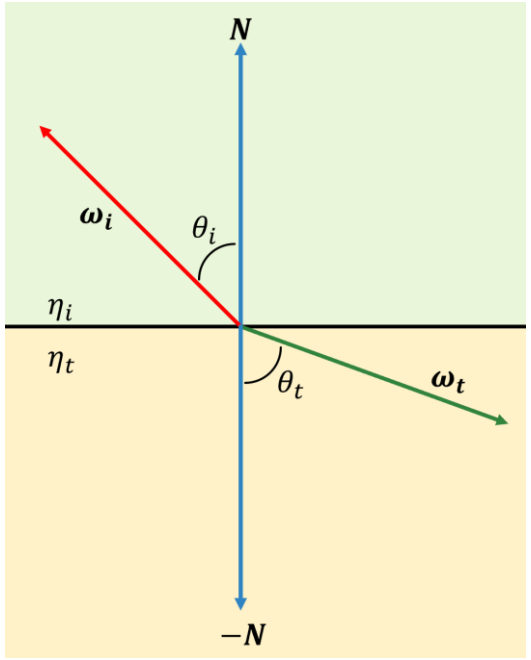


$$\omega_r = 2\,(\omega_i \cdot N)\,N - \omega_i$$



Specular Transmission

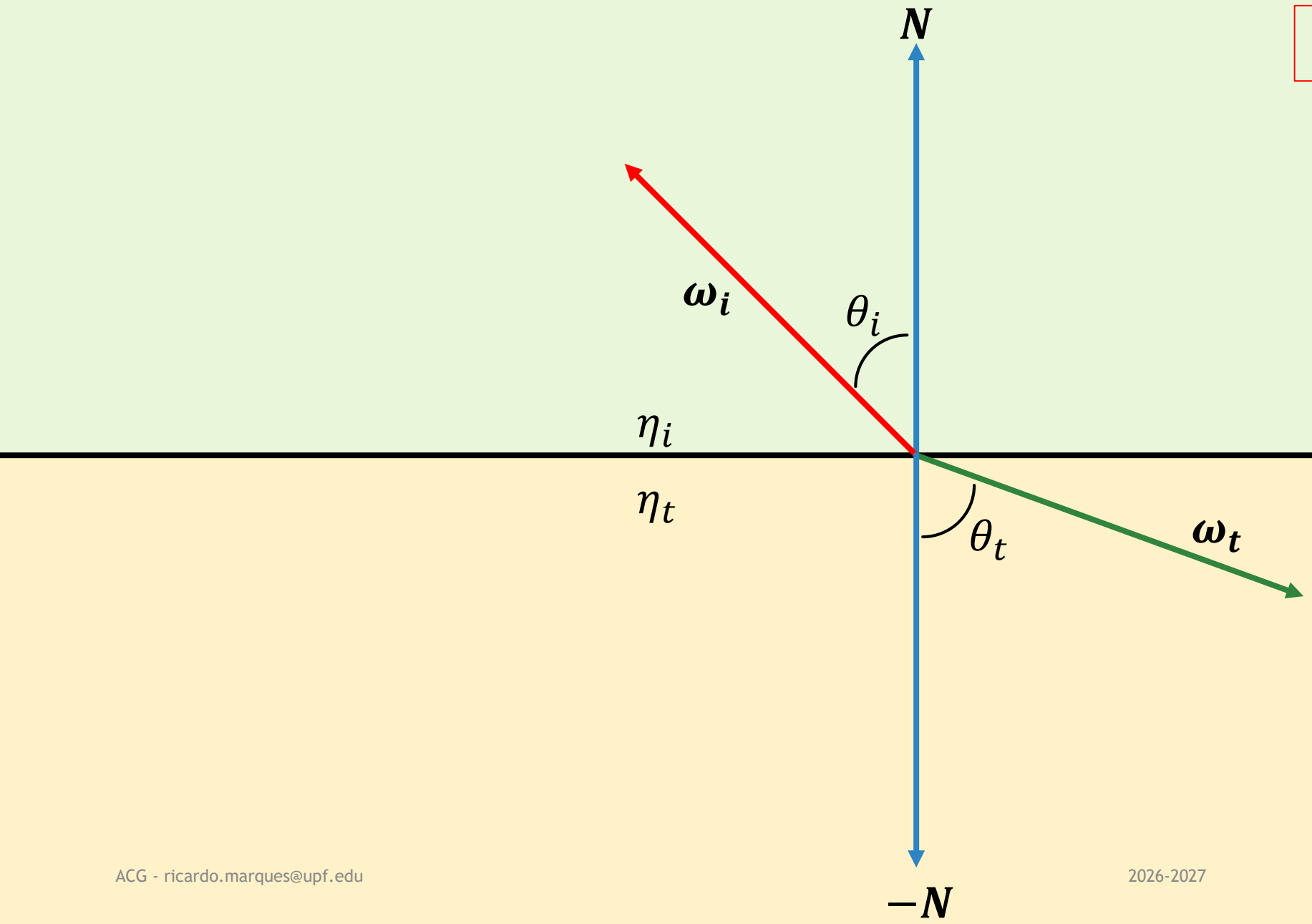
Specular Transmission: Problem Set-Up



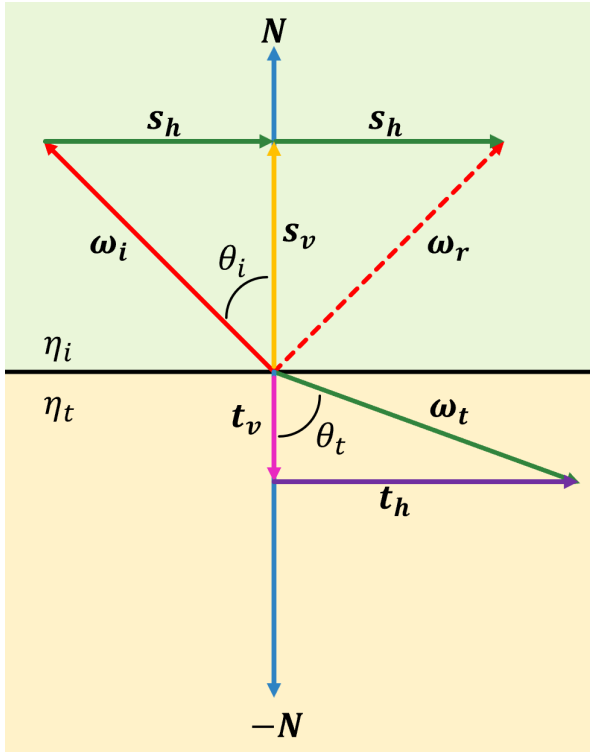
- Change of ray direction due to a change of propagating medium
- Problem set-up
  - $\boldsymbol{\omega}_i = -\boldsymbol{d}$, with $\boldsymbol{d}$ being the incident ray direction
  - η_i and η_o are the *index of refraction* (IOR) of the incident and outgoing propagating medium, respectively
  - $\boldsymbol{N}$ is the normal at the surface representing the interface between the two propagating media
- Objective: given the incident ray, compute the transmitted ray direction $\boldsymbol{\omega}_t$

Snell's law: $\frac{\sin\theta_i}{\sin\theta_t} = \frac{\eta_t}{\eta_i} = \eta$

Snell's law: $\frac{\sin\theta_i}{\sin\theta_t} = \frac{\eta_t}{\eta_i} = \eta$

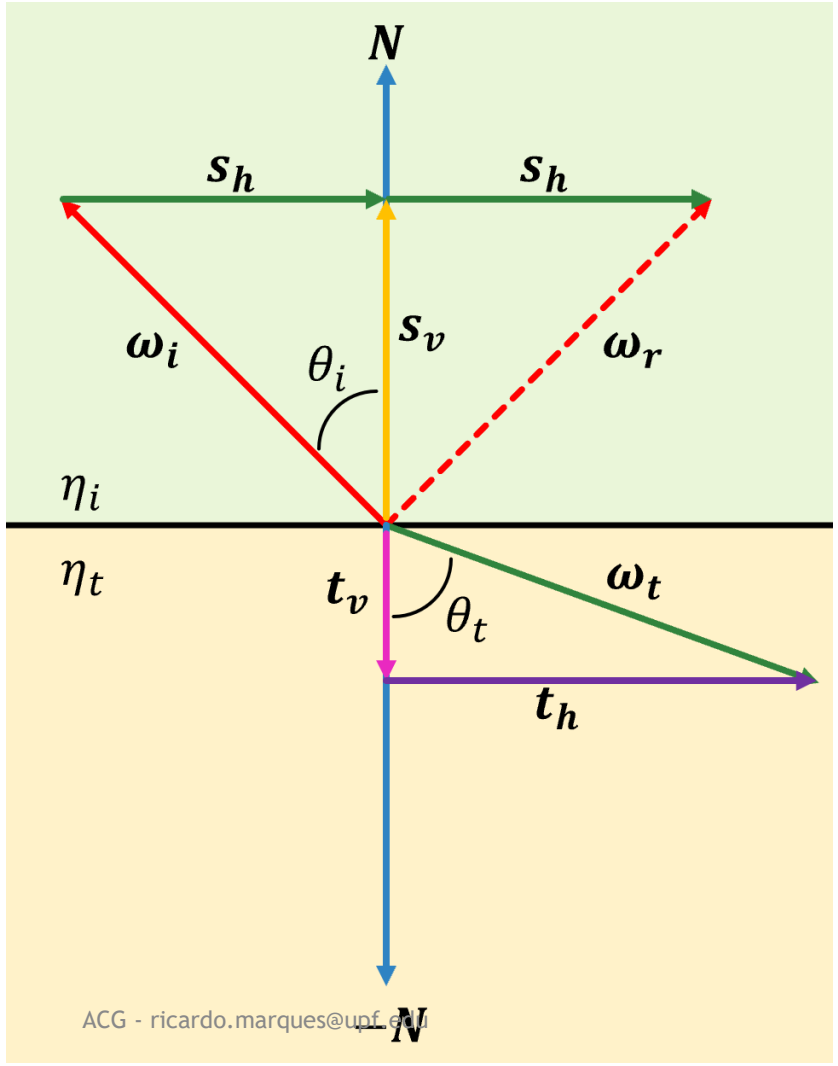


Specular Transmission: Strategy



- Use a similar strategy to that of the specular reflection
- Decompose the vector of interest into a vertical and a horizontal component
  - Reflection case: we used $\boldsymbol{\omega}_r = \boldsymbol{s}_v + \boldsymbol{s}_h$
  - Transmission case: we'll use $\boldsymbol{\omega}_t = \boldsymbol{t}_v + \boldsymbol{t}_h$
- Remarks
  - The direction of $\boldsymbol{t}_v$ is $-\boldsymbol{N}$ such that $\boldsymbol{t}_v = v\,(-\boldsymbol{N})$
  - The direction of $\boldsymbol{t}_h$ is $\boldsymbol{s}_h$ such that $\boldsymbol{t}_h = h\frac{\boldsymbol{s}_h}{\|\boldsymbol{s}_h\|}$
  - We only need to compute the scalars v and h

Computing ω_t

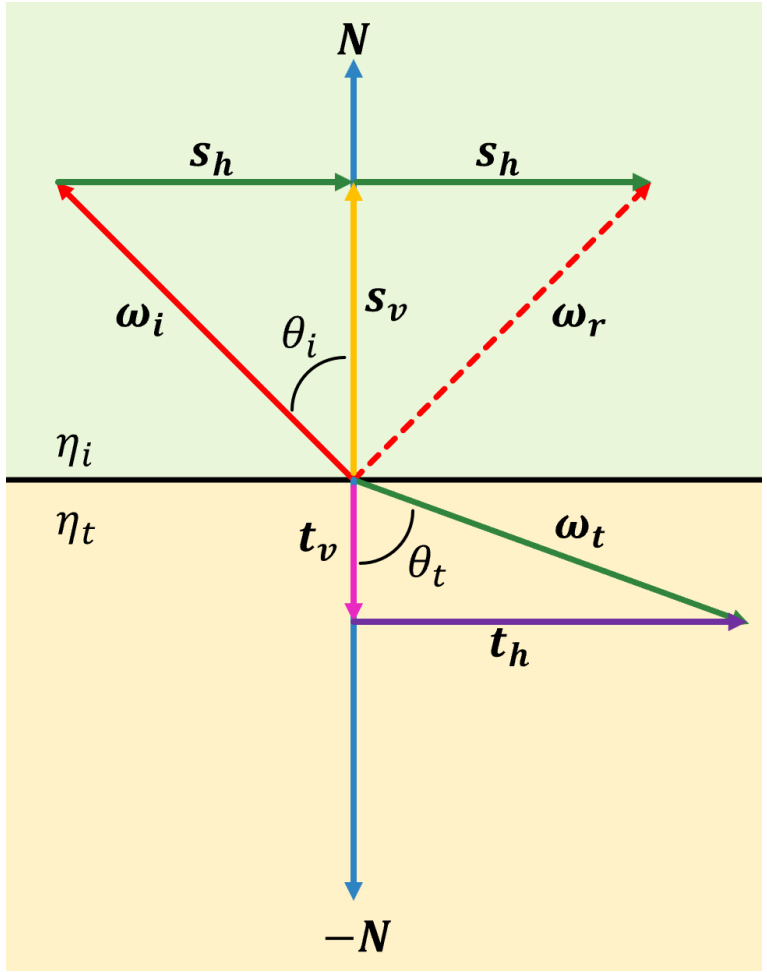


- Step 1: compute v such that $\boldsymbol{t}_v = v\,(-\boldsymbol{N})$
  - Because ω_t is normalized then $v = \cos\theta_t$
- Step 2: compute h such that $\boldsymbol{t}_h = h\frac{\boldsymbol{s}_h}{\|\boldsymbol{s}_h\|}$
  - Because ω_t is normalized then $h = \sin\theta_t$
- Putting it together
  - $\boldsymbol{\omega}_t = \boldsymbol{t}_v + \boldsymbol{t}_h = -N\cos\theta_t + \sin\theta_t\frac{\boldsymbol{s}_h}{\|\boldsymbol{s}_h\|}$
  - … but we don't know θ_t

Computing ω_t

$$\boldsymbol{\omega}_t = -\boldsymbol{N}\cos\theta_t + \sin\theta_t \frac{s_h}{\|s_h\|}$$

Snell's law: $\frac{\sin\theta_i}{\sin\theta_t} = \frac{\eta_t}{\eta_i} = \eta$



- Observation 1: $\cos\theta_t = \sqrt{1-\sin^2\theta_t}$. Using Snell's law ($\sin\theta_t = \eta^{-1}\sin\theta_i$) we can thus write:

$$-N\cos\theta_t = -N\sqrt{1-\eta^{-2}\sin^2\theta_i}$$

- Observation 2: $\|s_h\| = \sin\theta_i$. Using Snell's law, we can thus write:

$$\sin\theta_t \frac{s_h}{\|s_h\|} = \eta^{-1} s_h$$

- Observation 3: from the specular reflection, we know that:

$$\mathbf{s_h} = (\boldsymbol{\omega_i}\cdot\boldsymbol{N})\,\boldsymbol{N} - \boldsymbol{\omega_i}$$

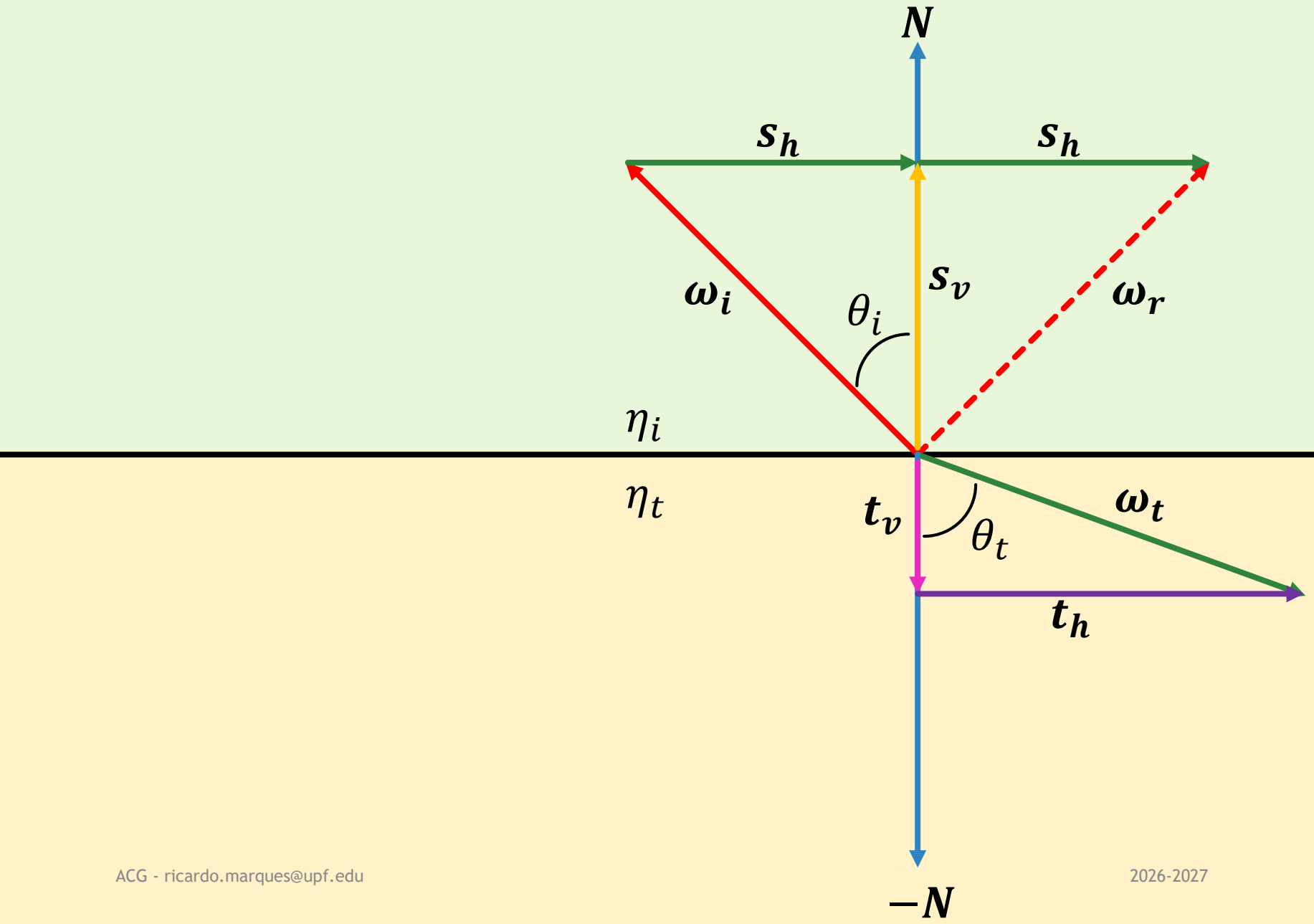
- Putting it all together and factoring-out $\boldsymbol{N}$, we have:

$$\boldsymbol{\omega}_t = \boldsymbol{N}\left(\eta^{-1}(\boldsymbol{\omega_i}\cdot\boldsymbol{N}) - \sqrt{1-\eta^{-2}\sin^2\theta_i}\right) - \eta^{-1}\boldsymbol{\omega_i}$$

where $\sin^2\theta_i$ can be easily computed as:

$$\sin^2\theta_i = 1-\cos^2\theta_i = 1-(\boldsymbol{\omega}_i\cdot\boldsymbol{N})^2$$

Note: it can happen that $\eta^{-2}\sin^2\theta_i > 1$. Then, the square root in the equation of ω_t cannot be computed (total internal reflection, no transmission, reflection towards ω_r instead).


N
s_h
s_h
s_v
ω_i
θ_i
ω_r
η_i
η_t
t_v
θ_t
ω_t
t_h
$-N$

Exercise 3: Reflection and Transmission

Consider a ray with origin $\mathrm{E} = (-3, -2, 3)$ which, after propagating through the air, intersects point $P = (0, 0.333, 1)$. Point P lies in the surface of an object made of water with normal $N = (-1, 0, 0)$. Considering that the refractive index of air is 1 and that the refractive index of water is 1.333, determine:

1) Whether the ray will be reflected or transmitted when hitting the surface at P

2) The reflected or refracted ray depending on the result obtained in 1)

Conclusion

Summary

- Ray-Scene Intersection Acceleration
  - Bounding Volumes, Ray Tracing Acceleration Structures
- Technical Insights on Ray Casting and Recursive Ray Tracing
  - Point Light Sources, Phong Model, Reflection, Refraction
- Key message
  - You now have the base for proceeding to a full implementation of RRT (Assignment 1 – Ray Tracing)
  - Probably need to come back to the (more technical) slides to fully assimilate its content
  - Totally empiric approach so far. Next classes we'll take a physically-based approach to rendering!

Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: Midterm	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: 1st Presentation	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: 2nd Presentation **(First Session)** **T12**: 2nd Presentation **(Second Session)**		**S5**: Work on FP	**D.FP**

Credits

- Thank you to Josep Blat for making available the slides for ACG (2023)
- Thank you to Beatriz Cabrero-Daniel (ex. GTI) for the video on Phong vs. Gouraud
- The Ray-Tracing Algorithms' slides and images are inspired on the NVidia's Ray-Tracing Essentials series (Eric Haines)
- Several slides and images on ray-object intersection are from Berkeley's CS184/284A (excellent) course on Computer Graphics



The End

Questions?